

Christopher Aykroyd

Game Programmer • Astrophysicist

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SUMMARY

Engineer and recent PhD graduate in astrophysics with a long-standing love for games, maths, and creative problem solving. I enjoy building systems that are technically challenging to develop and fun to interact with.

INTERESTS

Gameplay Graphics Procedural systems
AI systems Physics systems Scientific computing

PROGRAMMING

C# ●●●●● Python ●●●●●
Mathematica ●●●●● C++ ●●●●○
MATLAB ●●●●○ Java ●●●○○
C ●●○○○ JS/HTML/CSS ●●○○○

SOFTWARE & TOOLS

Unity Linux Git LaTeX PyTorch OpenCV

BASIC KNOWLEDGE

OpenGL Godot Unreal Maya Photoshop

AWARDS & GRANTS

CNES doctoral research grant. 2022–2025
USP best graduation project prize. 2021
PIBIC/CNPq undergraduate research grant. 2020–2021
Microsoft Imagine Cup French winner, Project Theatrall. 2018
Eiffel Grant of Excellence for undergraduate and master studies. 2017–2020

LANGUAGES

English C2 bilingual
Portuguese C2 bilingual
French C1 advanced
Spanish A2 basic

PUBLICATIONS

Non-conservative Hamiltonian perturbation methods for binary dynamics. Under review
C. Aykroyd, A. Bourgoïn, C. Le Poncin-Lafitte.

Hamiltonian treatment of non-conservative systems. *Phys. Rev. D*. 2026
C. Aykroyd, A. Bourgoïn, C. Le Poncin-Lafitte.

Hamiltonian normal forms for the post-Newtonian binary problem. *Phys. Rev. D*. 2025
C. Aykroyd, A. Bourgoïn, C. Le Poncin-Lafitte.

Embedded Star Catalog Calculation Tool for Autonomous Star Trackers. *PASP*. 2024
K. Bürger, F. Fialho, and C. Aykroyd.

Detection of magnetic galactic binaries in quasi-circular orbit with LISA. *Phys. Rev. D*. 2024
E. Savalle, A. Bourgoïn, C. Le Poncin-Lafitte, S. Mathis, M.-C. Angonin, and C. Aykroyd.

Secular dipole-dipole stability of magnetic binaries. *A&A*. 2023
C. Aykroyd, A. Bourgoïn, C. Le Poncin-Lafitte, S. Mathis, and M.-C. Angonin.

PROFESSIONAL EXPERIENCE

Doctoral Researcher 2022–2025
Observatoire de Paris • Astrophysics
Developed analytical and computational models for complex dynamical systems; published peer-reviewed research; built strong numerical, scientific and functional programming skills in Mathematica.

Research Intern 2019
Technicolor SA • Computer Graphics & Computer Vision
Worked on ageing regression of 3D facial meshes in PyTorch, graph-based generative models on surface data, and 3D face reconstruction from monocular video.

Game / Software Developer Intern 2018
Wildlife Studios • Unity / C#
Implemented enemy AI behaviour, metagame features and UI for a mobile FPS game.

EDUCATION

Ph.D. Astrophysics 2022–2025
Observatoire de Paris

M.Sc. Astrophysics 2021–2022
Observatoire de Paris
Highest honours

Engineering Degree 2017–2019
École Polytechnique
Focus on computer graphics, computer vision and AI; GPA: 3.52/4.00

B.Sc. Control Engineering 2014–2021
Universidade de São Paulo
Grade: 8.3/10; ranked 6/222

FEATURED TECHNICAL PROJECTS

Radiative Processes in Exoplanet Atmospheres 2021
Implemented Monte Carlo and backward ray-tracing simulations in C++ with an interactive visualization tool.

Möbius Voting for Surface Correspondence 2018–2019
Built a system for point-to-point correspondence between deformable 3D meshes, implemented in Python with a Three.js frontend.

Theatrall 2017–2018
Developed Node.js server-side and .NET application components for automatic subtitle generation for theatre plays; winner of the French finals of Microsoft Imagine Cup 2018.